

Aggregate Demand and Aggregate Supply

ECON 101: Macroeconomics

Exercise 1: Calculating AD and AS Equilibrium

The aggregate demand (AD) and short-run aggregate supply (SRAS) for an economy are given by the following equations:

$$AD : Y = 5000 - 200P$$

$$SRAS : Y = 2500 + 100P$$

where Y is real GDP and P is the price level.

Questions:

1. Find the equilibrium price level (P^*) and equilibrium output (Y^*).
2. Suppose consumer confidence decreases, shifting AD to $Y = 4800 - 200P$. Find the new equilibrium.
3. Illustrate the initial and new equilibrium graphically using AD and AS curves.

Exercise 2: Graphing the Effects of a Negative Supply Shock

Consider an economy in equilibrium at $Y = 3000$ and $P = 100$. A supply shock (e.g., an oil price increase) shifts SRAS leftward from:

$$SRAS_1 : Y = 3000 + 50P$$

$$SRAS_2 : Y = 2800 + 50P$$

Questions:

1. Determine the new equilibrium price level and output.
2. Illustrate the shift in SRAS using a graph.
3. What happens to inflation and unemployment after this shock?
4. If the central bank responds by increasing the money supply, how would AD shift?

Exercise 3: Fiscal Policy and AD Shifts

Suppose the government implements an expansionary fiscal policy by increasing spending by \$200 billion. The marginal propensity to consume (MPC) is 0.8.

Questions:

1. Calculate the total increase in GDP using the spending multiplier formula:

$$\text{Multiplier} = \frac{1}{1 - MPC}$$
$$\Delta Y = \text{Multiplier} \times \Delta G$$

2. If the initial AD curve was $Y = 4000 - 100P$, write the new AD equation after the fiscal policy.
3. Illustrate the impact on the AD-AS diagram, showing the shift in AD.
4. What happens to the unemployment rate if output increases?

Exercise 4: Long-Run Adjustment to a Recession

An economy is experiencing a **recessionary gap** with actual GDP at 2700 and full-employment GDP at 3000. Assume wages are **sticky in the short run** but adjust over time.

Questions:

1. Show the initial recessionary gap on an AD-AS diagram.
2. Over time, explain and graph how the SRAS curve adjusts to restore full employment.
3. What role does government intervention play in accelerating this adjustment?
4. How does this compare to classical vs. Keynesian views on policy?